

Harmonized Cross-Design Analytical Framework

Three independent open datasets are analyzed separately and integrated through evidence triangulation

Trupp 2022

Online quasi-randomized pre-post study
N = 84 adults

Exposure
Digital visual art viewing

Comparator
Active non-art cultural content

Primary estimand
Time by Condition anxiety contrast

Castellotti 2025

In-person uncontrolled field study
N = 92 adults

Exposure
Guided immersive exhibition visit

Temporal structure
Pre-visit and post-visit assessment

Primary estimand
Standardized within-person anxiety change

HEartS 2019

Cross-sectional online survey
N = 5,338 UK adults

Exposure
12-month visual-arts attendance frequency

Outcomes
Well-being, depression, loneliness,
and social connectedness

Primary estimand
Covariate-adjusted frequency associations

Construct-level harmonization with dataset-specific analysis

No individual-level pooling | Common effect direction | Design-specific estimands and uncertainty

Cross-design evidence triangulation

Direction | Magnitude | Precision | Robustness | Design-specific limitations

Integrated inference on immediate anxiety change, art specificity, and habitual visual-arts engagement